

Kelompok 3

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- Contoh Penggunaan Program, hasil eksekusi, dan hasil plot

The image displays two screenshots of a Jupyter Notebook environment, likely JupyterLab, showing the implementation and execution of the Newton-Raphson method for finding roots of a polynomial function.

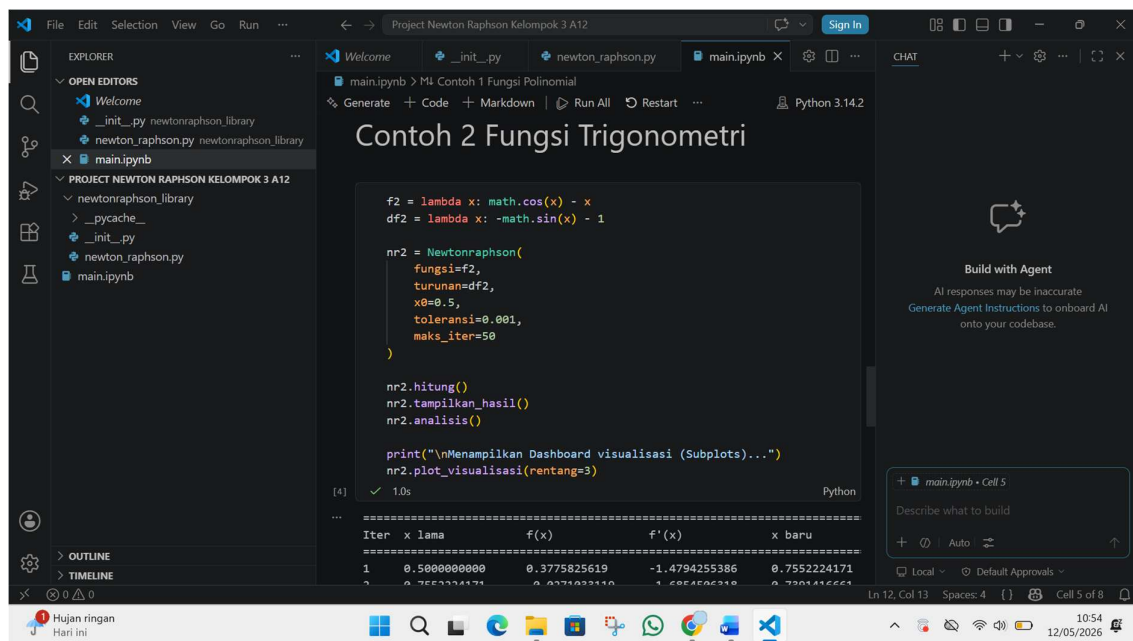
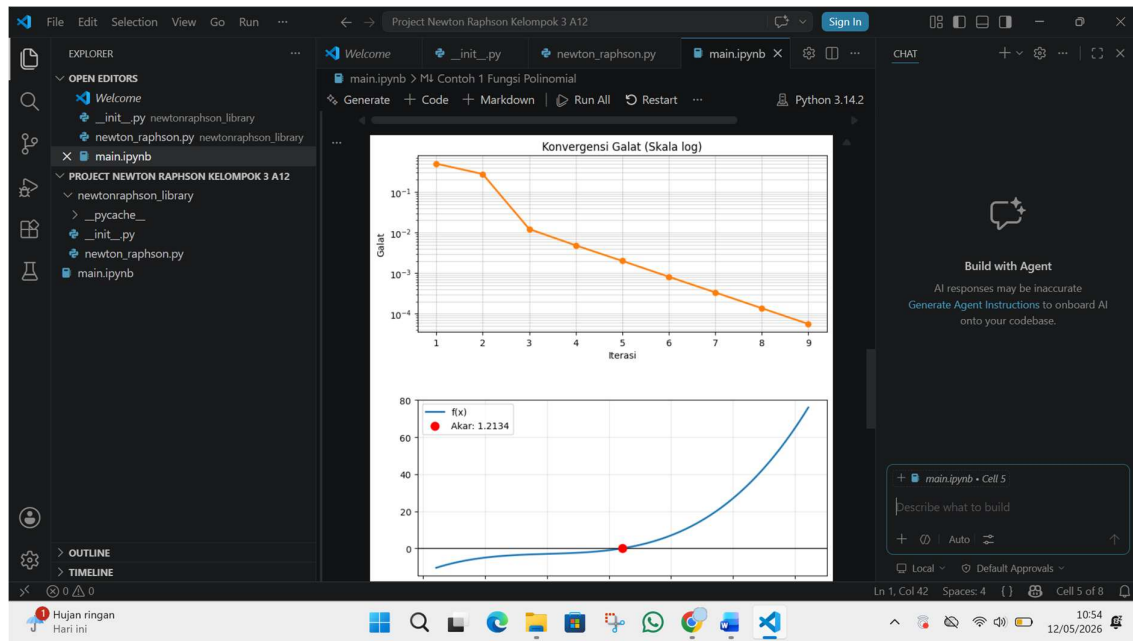
Top Screenshot: The notebook is titled "Contoh 1 Fungsi Polinomial". The code defines two functions: $f(x) = x^3 + x - 3$ and its derivative $f'(x) = 3x^2 + 2x - 3$. A `NewtonRaphson` class is defined with parameters for the function, derivative, initial guess ($x_0 = 1.0$), tolerance (0.0001), and maximum iterations (50). The class methods include `hitung` (calculate), `tampilkan_hasil` (display results), `analisis` (analysis), and `plot_visualisasi` (visualization). The execution output shows the first two iterations of the method:

Iter	x lama	f(x)	f'(x)	x baru
1	1.0000000000	-1.0000000000	2.0000000000	1.5000000000
2	1.5000000000	1.8750000000	6.7500000000	1.2222222222

Bottom Screenshot: This screenshot shows the continuation of the execution. The output displays the results of 9 iterations, indicating convergence. The final root found is $x = 1.2133953675$. The analysis section confirms convergence and provides details about the initial guess, iterations, tolerance, and final error.

==== Analisis Konvergensi====
Metode Newton-Raphson berhasil konvergen.
Tebakan awal : 1.0
Akar hampiran : 1.2133953675
Jumlah Iterasi : 9
Toleransi : 0.0001
Galat terakhir : 5.6120898454e-05

Menampilkan Dashboard Visualisasi (Subplots)...



Project Newton Raphson Kelompok 3 A12

main.ipynb > M1 Contoh 1 Fungsi Polinomial

Generate + Code + Markdown Run All Restart Python 3.14.2

```
Iter x lama f(x) f'(x) x baru
=====
1 0.500000000 0.3775825619 -1.4794255386 0.7552224171
2 0.7552224171 -0.0271033119 -1.6854506318 0.7391416661
3 0.7391416661 -0.0000946154 -1.6736538108 0.7390851339
=====
Status : konvergen
Akar ditemukan : x = 0.7390851339
Jumlah iterasi : 3

=== Analisis Konvergensi===
Metode Newton-Raphson berhasil konvergen.
Tebakan awal : 0.5
Akar hampiran : 0.7390851339
Jumlah Iterasi : 3
Toleransi : 0.001
Galat terakhir : 5.6532229072e-05

Menampilkan Dashboard visualisasi (Subplots)...
```

Konvergensi Galat (Skala log)

Build with Agent

AI responses may be inaccurate. Generate Agent Instructions to onboard AI onto your codebase.

main.ipynb • Cell 5

S

Local Default Approvals

Ln 12, Col 13 Spaces 4 Cell 5 of 8

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Project Newton Raphson Kelompok 3 A12

main.ipynb > M1 Contoh 1 Fungsi Polinomial

Generate + Code + Markdown Run All Restart Python 3.14.2

Konvergensi Galat (Skala log)

Galat

Iterasi

f(x)

Akar: 0.7391

Build with Agent

AI responses may be inaccurate. Generate Agent Instructions to onboard AI onto your codebase.

main.ipynb • Cell 5

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Local Default Approvals

Ln 12, Col 13 Spaces 4 Cell 5 of 8

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